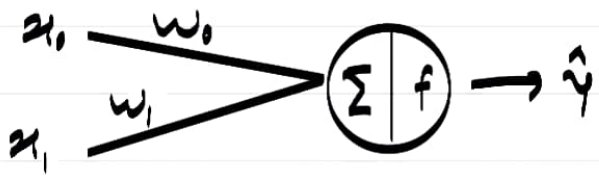
Deep Learning

29-05-26

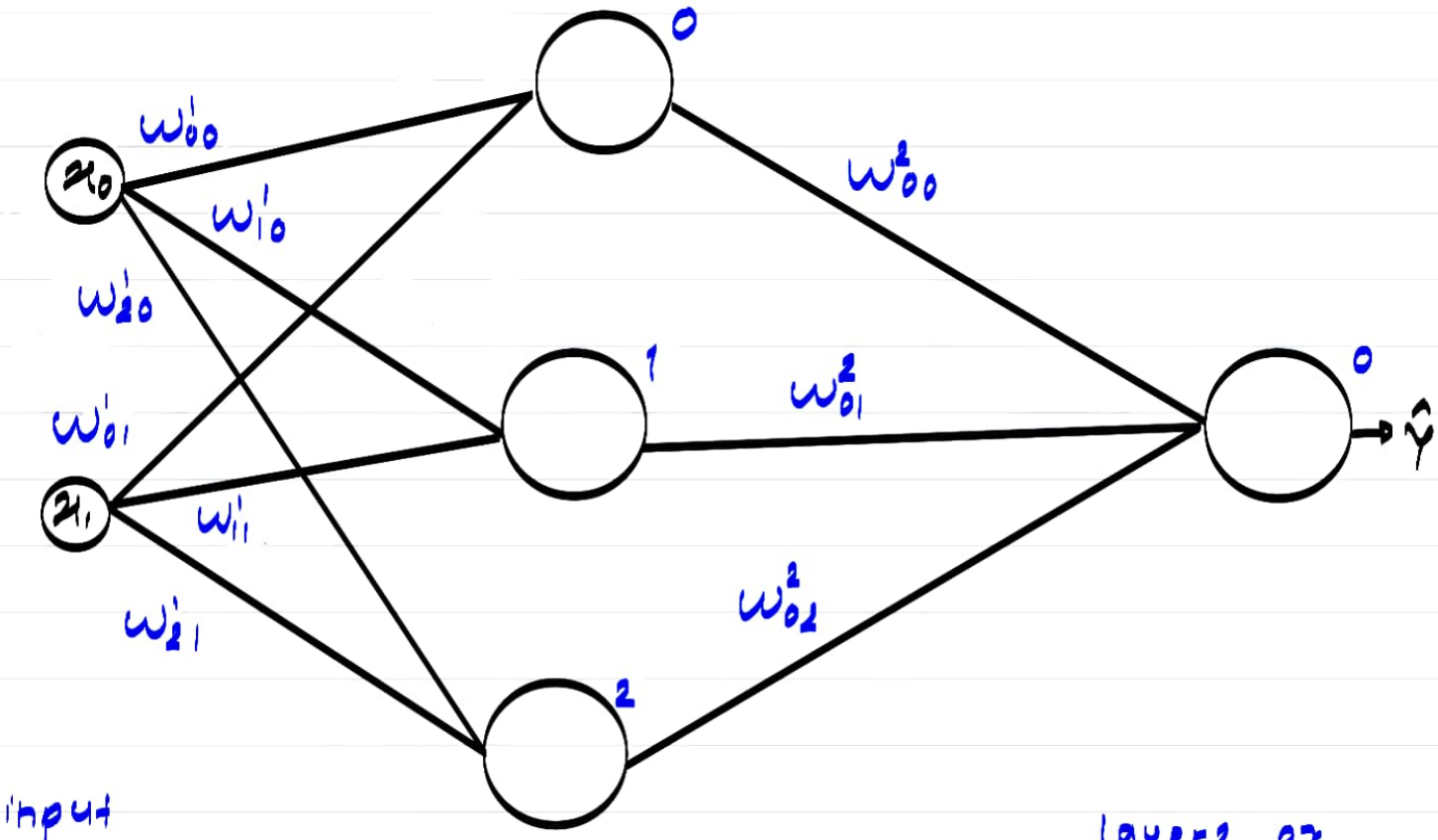
ANN: Artificial Neural Network

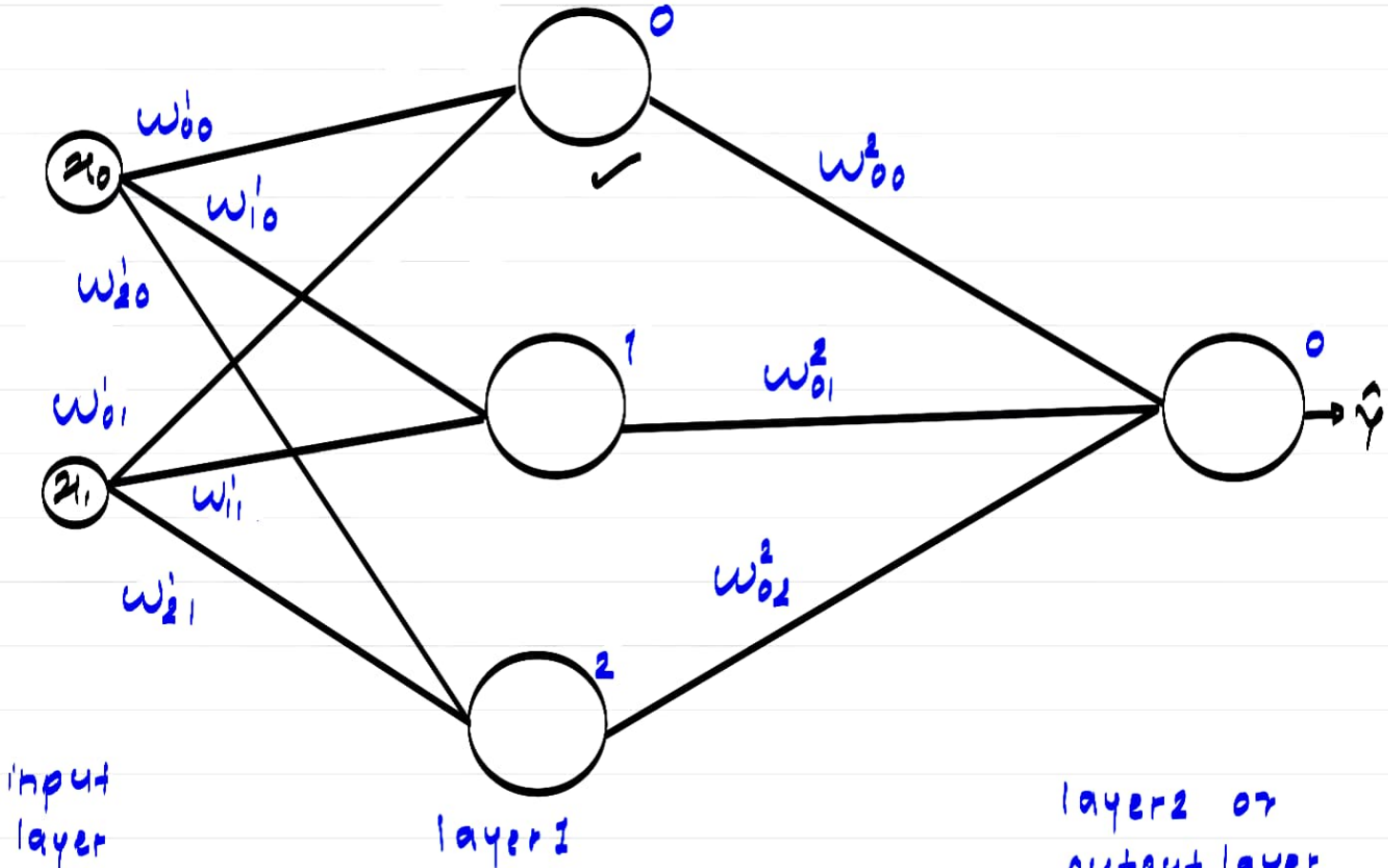
- when multiple perceptrons are combined through multiple layers to form a complex network then its called ANN model

single perceptron



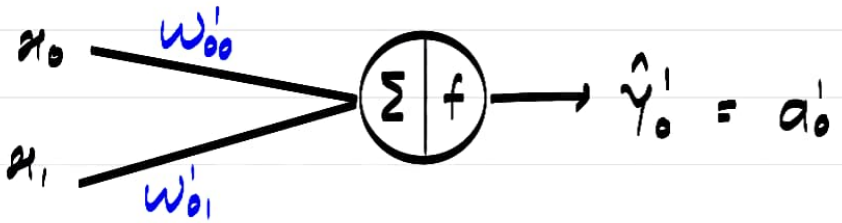
ANN





Forward propagation in ANN :

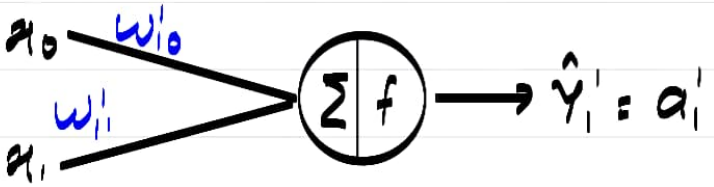
node 0 / neuron 0 in layer 1



$$z^1_0 = w^1_{00} x_0 + w^1_{01} x_1$$

$$\hat{y}^1_0 = f(z^1_0) = a^1_0$$

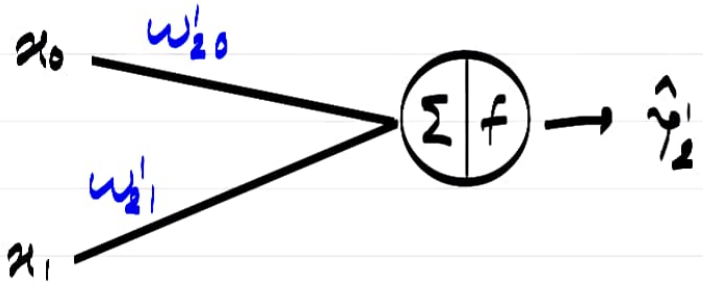
node 1 / neron 1 in layer 1



$$z^1_1 = w^1_{10} x_0 + w^1_{11} x_1$$

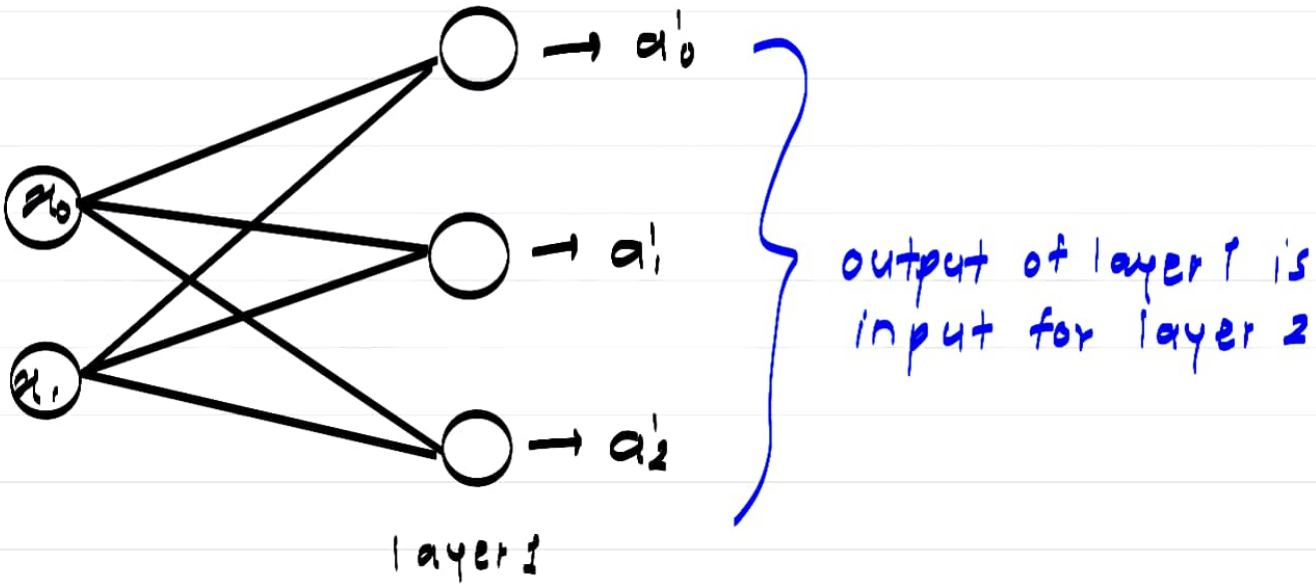
$$\hat{y}^1_1 = f(z^1_1) = a^1_1$$

node 2 / neuron 2 in layer 1

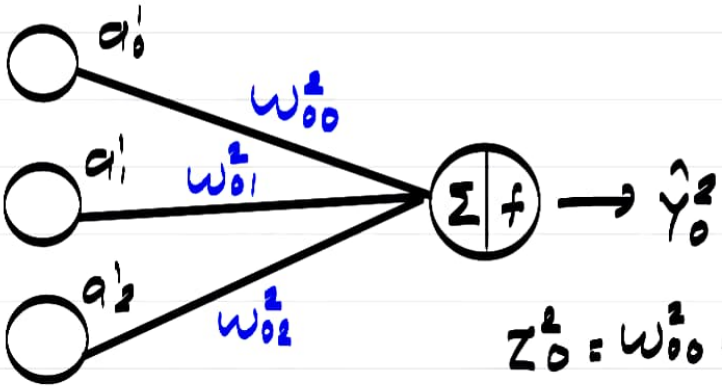


$$z^1_2 = w^1_{20} x_0 + w^1_{21} x_1$$

$$\hat{y}^1_2 = f(z^1_2) = a^1_2$$



node 0 / neuron 0 in layer 2 / output layer



$$z^2_0 = w^2_{00} a^1_0 + w^2_{01} a^1_1 + w^2_{02} a^1_2$$

$$\hat{y}^2_0 = f(z^2_0) = a^2_0$$

As layer 2 is also the last and the output layer, $\hat{y}_0^2 = a_0^2$ will be considered as model prediction

Naming conventions in ANN

- ANN has multiple layer
- Each layer has multiple nodes/neurons
- Each neuron in given layer takes input from all the neurons present in its previous layer
- Each connection between neurons has weight associated to it.

W_{ij}^l

$l \rightarrow$ layer l

$i \rightarrow$ neuron i in layer l

$j \rightarrow$ neuron j in layer $l-1$

$W_{ij}^l \rightarrow$ weight associated to connection betn neuron i in layer l and neuron j in layer $l-1$

$W_{02}^1 \rightarrow$ layer 1
neuron 0 in layer 1
neuron 2 in layer 0 / input layer

$W_{12}^2 \rightarrow$ layer 2
neuron 1 in layer 2
neuron 2 in layer 1

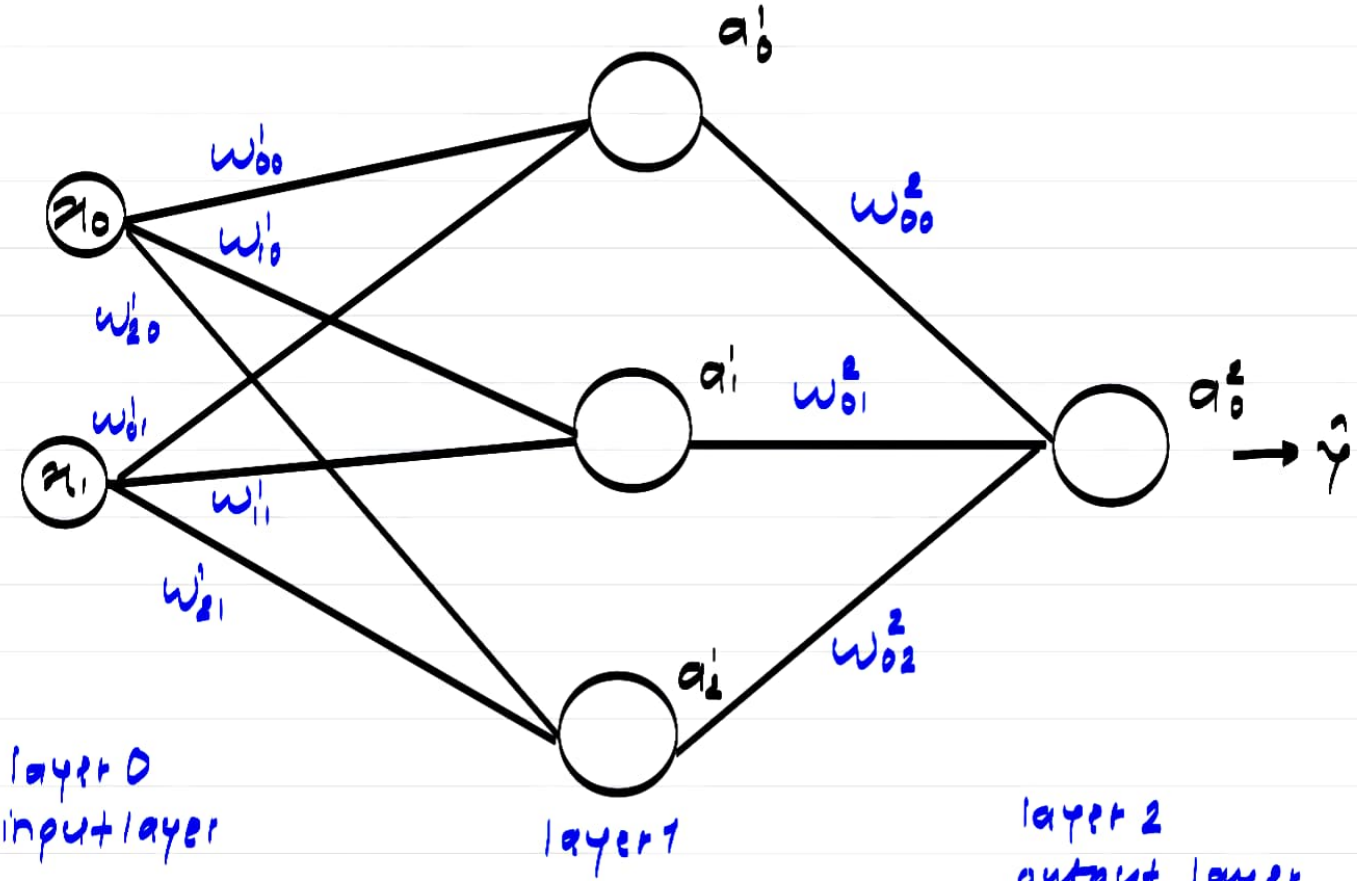
z_i^l weighted aggregation computed in neuron i in layer l

z_1^1 → weighted agg computed in neuron 1 in layer 1

a_i^l activation computed in neuron i in layer l

$a_i^l = f(z_i^l)$

$a_1^1 = f(z_1^1)$ $\quad$ $a_3^2 = f(z_3^2)$



node 0 in layer 1

$$z_0^1 = w_{00}^1 x_0 + w_{01}^1 x_1$$
$$a_0^1 = f^1(z_0^1)$$

node 1 in layer 1

$$z_1^1 = w_{10}^1 x_0 + w_{11}^1 x_1$$
$$a_1^1 = f^1(z_1^1)$$

node 2 in layer 1

$$z_2^1 = w_{20}^1 x_0 + w_{21}^1 x_1$$
$$a_2^1 = f^1(z_2^1)$$

All neurons in same layer always have same activation function

Layer 1 operations using matrix multiplication:

$$W^1 = \begin{bmatrix} w_{00}^1 & w_{01}^1 \\ w_{10}^1 & w_{11}^1 \\ w_{20}^1 & w_{21}^1 \end{bmatrix} \qquad X = \begin{bmatrix} x_{00} & x_{10} \\ x_{01} & x_{11} \end{bmatrix} \begin{matrix} \downarrow & \downarrow \\ d_0 & d_1 \end{matrix}$$

we have 2 datapoints

$d_0 : (x_{00} \quad x_{01})$

$d_1 : (x_{10} \quad x_{11})$

$$Z^1 = W^1 X$$

$$Z^1 = W^1 X \qquad X = A^0$$

$$Z^1 = W^1 A^0$$

$$W^1 = \begin{bmatrix} w^1_{00} & w^1_{01} \\ w^1_{10} & w^1_{11} \\ w^1_{20} & w^1_{21} \end{bmatrix} \begin{array}{l} \rightarrow \text{weights for node 0 in layer 1} \\ \rightarrow \text{weights for node 1 in layer 1} \\ \rightarrow \text{weights for node 2 in layer 1} \end{array}$$

$$Z^1 = W^1 A^0 = W^1 X$$

$$Z^1 = \begin{bmatrix} w^1_{00} & w^1_{01} \\ w^1_{10} & w^1_{11} \\ w^1_{20} & w^1_{21} \end{bmatrix} \begin{bmatrix} x_{00} & x_{10} \\ x_{01} & x_{11} \end{bmatrix}$$

3×2 $\qquad$ 2×2

now put

$$w^1_0 = [w^1_{00} \;\; w^1_{01}]$$
$$w^1_1 = [w^1_{10} \;\; w^1_{11}]$$
$$w^1_2 = [w^1_{20} \;\; w^1_{21}]$$

$$x_0 = \begin{bmatrix} x_{00} \\ x_{01} \end{bmatrix} \qquad x_1 = \begin{bmatrix} x_{10} \\ x_{11} \end{bmatrix}$$

$$Z^1 = \begin{bmatrix} w^1_0 \\ w^1_1 \\ w^1_2 \end{bmatrix} \begin{bmatrix} x_0 & x_1 \end{bmatrix}$$

$$Z^1 = \begin{bmatrix} w^1_0 \cdot x_0 & w^1_0 \cdot x_1 \\ w^1_1 \cdot x_0 & w^1_1 \cdot x_1 \\ w^1_2 \cdot x_0 & w^1_2 \cdot x_1 \end{bmatrix}$$

$$Z^1 = \begin{bmatrix} w^1_{00} x_{00} + w^1_{01} x_{01} & w^1_{00} x_{10} + w^1_{01} x_{11} \\ w^1_{10} x_{00} + w^1_{11} x_{01} & w^1_{10} x_{10} + w^1_{11} x_{11} \\ w^1_{20} x_{00} + w^1_{21} x_{01} & w^1_{20} x_{10} + w^1_{21} x_{11} \end{bmatrix} = \begin{bmatrix} z^1_{00} & z^1_{01} \\ z^1_{10} & z^1_{11} \\ z^1_{20} & z^1_{21} \end{bmatrix}$$

$$Z^1 = W^1 X$$

$$Z^1 = \begin{bmatrix} z^1_{00} & z^1_{01} \\ z^1_{10} & z^1_{11} \\ z^1_{20} & z^1_{21} \end{bmatrix}$$

z^l_{ij} → weighted agg in layer l inside node i in layer l for datapoint j

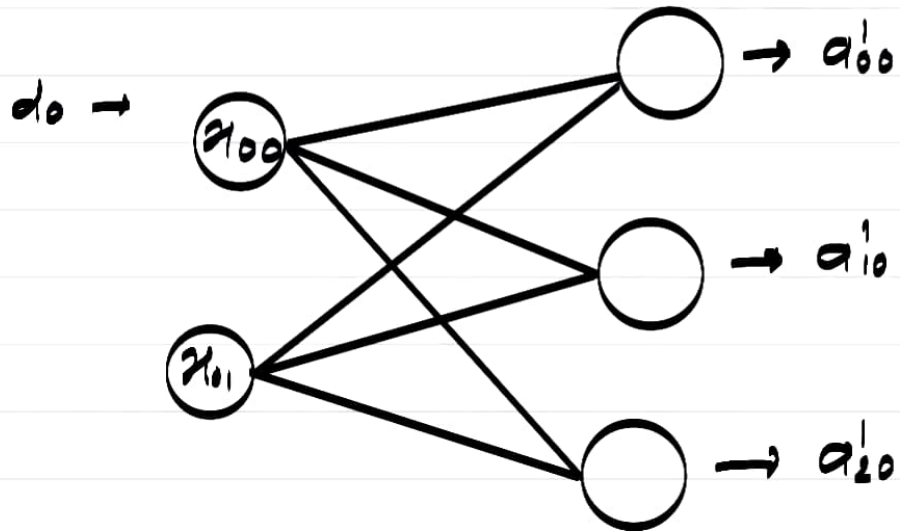
z^1_{10} → weighted agg in layer 1 inside node 1 in layer 1 for datapoint d_0

$$A^1 = f(Z^1)$$

$$A^1 = \begin{bmatrix} f(z^1_{00}) & f(z^1_{01}) \\ f(z^1_{10}) & f(z^1_{11}) \\ f(z^1_{20}) & f(z^1_{21}) \end{bmatrix} = \begin{bmatrix} a^1_{00} & a^1_{01} \\ a^1_{10} & a^1_{11} \\ a^1_{20} & a^1_{21} \end{bmatrix}$$

$a^1_{00}, a^1_{10}, a^1_{20}$ → for d_0

$a^1_{01}, a^1_{11}, a^1_{21}$ → for d_1



Forward propagation in layer 1

layer 1 gets input from layer 0 as $X = A^0$

layer 1 has weight matrix W^1

$$Z^1 = W^1 A^0$$

$$A^1 = f^1(Z^1)$$

Forward propagation in layer 2

layer 2 gets input from layer 1 as A^1

layer 2 has weight matrix W^2

$$Z^2 = W^2 A^1$$

$$A^2 = f^2(Z^2)$$

Forward propagation for layer l

layer l gets inputs from layer l-1 as A^{l-1}

layer l has weight matrix W^l

$$Z^l = W^l A^{l-1}$$

$$A^l = f(Z^l)$$

In forward pass, 2 steps occur in every layer

1) weighted aggregation of inputs

2) Apply activation function

Using weighted agg. ANN tries to learn importance of inputs in every layer and relationship between input and output

Activation function applied after weighted agg helps further in learning complex relationships between input and output